DATA ANALYST PORTFOLIO PROJECT · THE STORY BEHIND IT

The Data I Had, But Didn't Know How to Use

Sun Sahar · Companion to the presentation and the SQL file

The presentation shows the findings and the SQL file shows how they were produced. This page gives the context behind them: the business, the decisions I made at the time, why I went back to the data, and what changed in how I think.

1. The business

I ran a small trading-education business. Its model was simple. A free course, promoted with Meta ads (Facebook and Instagram), brought people in. Each person who signed up became a lead, and we tried to reach each lead by phone to offer the full course with personal mentorship. People became paying customers through several routes: word of mouth, ad campaigns, webinars and personal calls.

I ran the advertising myself across two ad accounts from mid-2024 to early 2026. I could see how many leads came in and what each lead cost, and that is what I steered by. What happened after the lead was harder. Getting people from a form to a sales call was the hardest part of the business, and it was also the part I measured least.

2. What I did at the time

- **2024: finding the message.** We spent little. We did not yet know what to advertise, so we put our effort into understanding our customers rather than into ads.
- **February – August 2025: lead campaigns.** Free-course campaigns ran through Meta's built-in sign-up form. March was the busiest month, and calling the new leads kept us fully occupied. In July I tried a landing page that offered a free personal fit meeting instead.
- **Late 2025: new offers.** A condensed ₪97 course, webinars, messaging campaigns, and exposure campaigns to grow the Instagram audience. The mini-course did not sell.

Throughout this period I judged results mainly by lead count and cost per lead. I changed creatives, budgets, prices and the sales process, and I mainly relied on those two numbers to judge the changes.

3. Why I went back to the data

As a Data Analyst student I wanted a project that was real, not a tutorial dataset. My own business had about ₪39,000 of advertising history, and I had made real decisions based on it. That gave me something rare: I could compare what the data said with what I had actually done and believed at the time. The question I set myself was: *what could this data have told me then, and what could it never tell me?*

4. How the data changed my understanding

> - **I was looking at the wrong step.** When costs rose I suspected the form and the offer. Most of the change happened before anyone clicked.
> - **The step that paid was the one I measured least.** Meta counted every lead automatically. Calls, meetings and sales were barely recorded, and the little I had pointed to a bottleneck after the lead.
> - **I never recorded where my paying customers came from.** When I added my customers' payments, I could see that payments were higher while the lead campaigns ran, but sales were happening before the ads too. Without a channel on each customer, I cannot say which ones the ads brought, and that is the most valuable information I did not keep.
> - **Some of my own memories were wrong.** I remembered every campaign as a lead campaign. The data showed that almost a quarter of the budget went to exposure and to offer tests, and that those tests were judged by clicks and views, not by sales.

5. How I worked

- **Understand the data before counting.** The ad export held a row for every ad on every day, whether or not it ran. About 99% of the rows were empty padding, so the first rule was that only days with spend count.
- **Check before trusting.** The key reported totals (spend, leads, active days and months) were cross-checked between Python and BigQuery before any analysis began. During validation, I found a discrepancy between an earlier business record indicating 46 students and the payment dataset containing 38 distinct paying customers across 50 payments (₪142,234, excluding VAT). Since the two figures could not be reliably reconciled, I used the 38 customers supported by the payment records for the financial analysis.
- **Keep memory and data apart.** I wrote down what I remembered in a separate document. Where the two disagreed, I based the analysis on the data.
- **Protect the people behind the data.** Customer names were replaced by anonymous IDs and all notes were removed before the payments were used.
- **Say only what the data supports.** No payment is linked to an ad, so there is no customer-acquisition cost, return on ad spend or profit figure. Comparisons are labelled as comparisons, and the what-if plan is labelled as a plan, not a forecast.

6. What I would measure differently

- Tag every lead and every paying customer with its source: ad, webinar, referral or call.
- Record each lead's journey: called, meeting scheduled, meeting held, sale, amount.
- Give every creative a unique name, so creatives can be compared.
- Judge every test by the outcome that pays, and steer by cost per scheduled meeting rather than cost per lead.

7. How this project was built

I built this project with AI tools. I did not write the SQL or Python code from scratch: AI tools wrote it. My part was to supply the business context and the data, to define and refine the questions, to run the queries in BigQuery, to build the dashboard with guidance, to challenge results that did not match what I knew about the business, and to interpret what the findings mean. I am still learning to write this code on my own, and every query is documented so that I can.

What I am sending

Presentation (PDF)	The business story and findings in nine slides, with the what-if plan and an appendix on method and metric definitions. It includes the link to the live Looker Studio dashboard.
SQL file	Every query behind the findings, in story order, each with a short header: the question, the method, the result and its limits.
Python code (GitHub)	The cleaning, anonymisation and check scripts: github.com/Ssahar6/python
This document	The context: why the project exists and how I approached it.

Sun Sahar · All data is my own business's data: aggregate advertising figures and anonymised payments (excl. VAT). No customer names or contact details appear anywhere in the project.